



SOUTHEAST UNIVERSITY

Assignment on Static Routing Configuration in a multi-Router Network.
Course Title: Communication theory LAB
Course Code: ETE282.8

Submitted by

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Date of Submission
27.04.2026

☐ Experiment No: 04

☐ Experiment Name: Static Routing Configuration in a multi-Router Network.

☐ Introduction: In modern enterprise and campus-level network environments data must often traverse multiple network segments to reach its destination. Static Routing is one of the most foundational techniques for enabling this inter-network communication. Unlike dynamic routing protocols, static routing requires a network administrator to manually define the precise paths that packets must follow to reach remote networks.

In this LAB Experiment we build a multi-router topology was designed and implemented using Cisco Packet Tracer. Three routers [Router0, Router1 and Router2] were interconnected using serial and fast-Ethernet links forming a triangle of WAN connections. Multiple end-devices are attached to the router via switches. Static routes were then manually configured on each router to ensure full end-to-end reachability across all network segments. The Experiment reinforces the understanding of IP-Routing fundamentals, network address planning

and the administrative Process of static Route Configuration. Let's go through the following Processes.

Objectives:

- Design and Implement a multi-~~lan~~ router topology using Cisco Packet Tracer.
- Assign and Configure distinct IP address ranges for each network segment.
- Configure static routes on each router to enable inter network Packet forwarding.
- Verify end-to-end Connectivity across all network segments using ICMP Ping tests.
- Understand the Role of Serial and FastEthernet interfaces in WAN and LAN Connectivity.
- Differentiate between directly Connected networks and remotely reachable networks.

Theoretical Background:

1. Static Routing: Static Routing is the Process by which an administrator manually defines routes in a router's routing table. Each static route entry specifies a destination network, a subnet mask and a next-hop IP address [or exit interface].

The router then uses this information to forward packets toward the correct destination. Static routes are predictable and secure but do not adapt automatically to network topology changes - making them best suited for small, stable networks or specific policy-based routing scenarios.

2. IP Routing and the Routing Table: A router routing table is a data structure that stores mappings of destination networks to outgoing interfaces or next hop addresses. When a packet arrives at a router, the router performs a longest-prefix match in against its routing table to determine where to forward the packet. Directly connected networks are automatically added to the table, while static or dynamically learned routes must be configured explicitly.

3. Serial Links and WAN Connectivity: Serial interfaces [se 2/0, se 3/0] are used to simulate WAN point-to-point connections between routers. In Cisco Packet Tracer, serial links require a clock rate on the DCE [Data Communication Equipment] end to establish synchronization. Fast Ethernet interfaces (fa 0/0, fa 0/1) are used for LAN-side connectivity to switches and end devices.

4. Network Segmentation: Each Router Interface that connects to a different Physical segment must reside in a unique Subnet. This ensures that broadcasts remain contained within their respective network segments and that routing decisions can be made unambiguously. In this topology, Each LAN network and each WAN link between routers was assigned a dedicated address block.

Network Topology and Hardware Inventory:

1. Hardware Utilized:

- 3x Cisco PT-Routers [Router0, Router1, Router2]
- 1x Cisco 2960-24TT Switch
- 2x PCs [PC0, PC1]
- 3x Laptops [Laptop0, Laptop1, Laptop2]
- 2x Servers [Server0, Server1]

2. Network segments and IP Address Allocation:

- In this Network we are using All Class A Net ID's and Subnet mask 255.0.0.0. In Below all the IP address allocation will be showed.

Network Segment	Network Address	Connected Devices
Router0 LAN (Fa0/0)	10.0.0.0/8	PC0
Router0 LAN (Fa1/0)	11.0.0.0/8	Laptop0
Router0 - Router1 WAN	17.0.0.0/8	Se 2/0 - Se 2/0
Router0 - Router2 WAN	12.0.0.0/8	Se 3/0 - Se 2/0
Router1 - Router2 WAN	14.0.0.0/8	Se 3/0 - Se 3/0
Router1 LAN (Fa0/0)	15.0.0.0/8	Laptop2
Router1 LAN (Fa1/0)	16.0.0.0/8	Server1
Router2 LAN (Fa0/0)	13.0.0.0/8	Switch → PC1, Laptop1, Server2

Implementation and Configuration steps:

1. Router Interface Configuration: Each Router's Fast-Ethernet Interfaces were configured with an IP address to act as the default gateway for their respective LAN segments. Serial Interfaces were configured on both ends of each WAN Link with Clock rate set on the DCE end. The following CLI Commands were applied on each Router:

Router0: → Interface Fa0/0
 IP address 10.0.0.1 255.0.0.0
 no shutdown
 Interface Fa1/0
 IP address 11.0.0.1 255.0.0.0
 no shutdown

- Interface Se2/0
ip address 17.0.0.1 255.0.0.0 Clock rate 9600
no shutdown
- Interface Se3/0
ip address 12.0.0.1 255.0.0.0
no shutdown

Router-1:

- Interface Fa0/0
ip address 15.0.0.1 255.0.0.0
no shutdown
- Interface Fa1/0
ip address 16.0.0.1 255.0.0.0
no shutdown
- Interface Se2/0
ip address 17.0.0.2 255.0.0.0
no shutdown
- Interface Se3/0
ip address 14.0.0.1 255.0.0.0 Clock rate 9600
no shutdown

Router-2:

- Interface Fa0/0
ip address 13.0.0.2 255.0.0.0
no shutdown
- Interface Se2/0
ip address 12.0.0.2 255.0.0.0
no shutdown
- Interface Se3/0
ip address 14.0.0.2 255.0.0.0
no shutdown

2. Static Route Configuration: After all interfaces were brought up static routes were added to each router's routing table for every remote network. The general syntax used was:

→ IP route <destination_network> <subnet_mask> <next_hop_IP>

Static routes on Router-0:

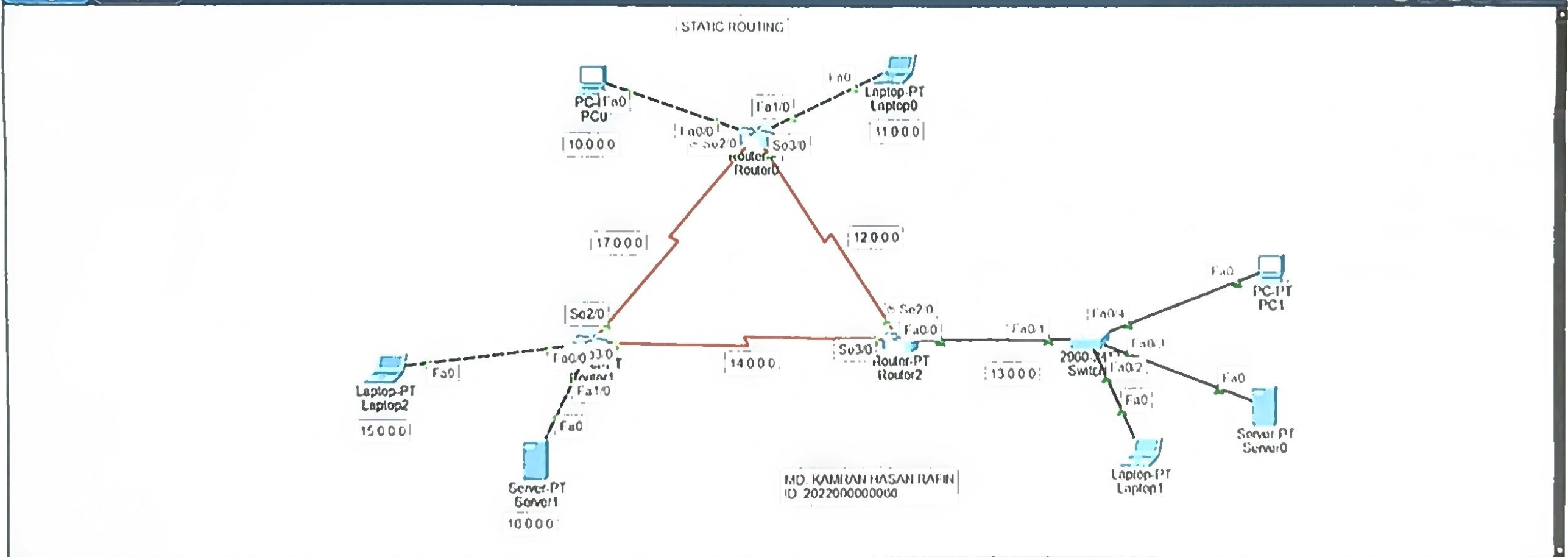
→ IP route 13.0.0.0 255.0.0.0 serial 3/0
→ IP route 14.0.0.0 255.0.0.0 serial 3/0
→ IP route 15.0.0.0 255.0.0.0 serial 3/0
→ IP route 16.0.0.0 255.0.0.0 serial 3/0

Static routes on Router-1:

→ IP route 10.0.0.0 255.0.0.0 serial 2/0
→ IP route 11.0.0.0 255.0.0.0 serial 2/0
→ IP route 12.0.0.0 255.0.0.0 serial 2/0
→ IP route 13.0.0.0 255.0.0.0 serial 3/0

Static routes on Router-2:

→ IP route 10.0.0.0 255.0.0.0 serial 2/0
→ IP route 11.0.0.0 255.0.0.0 serial 2/0
→ IP route 15.0.0.0 255.0.0.0 serial 3/0
→ IP route 16.0.0.0 255.0.0.0 serial 3/0
→ IP route 17.0.0.0 255.0.0.0 serial 3/0



3. End-Device Configuration: All PCs and Laptops were manually assigned static IP addresses within their respective network segments with their default gateway set to the IP address of the connected router interface. The switch on Router-2 LAN segment was used to extend connectivity to multiple devices including PC1, Laptop1 and Server-0.

Result and Verification: Following the complete of all router interfaces and static routes, end-to-end connectivity tests were performed using ICMP echo requests (Ping). The results were observed via the Cisco Packet Tracer Simulation Panel and the PDU List window:

Source	Destination	Type	Time (s)	Status
Laptop 0	Laptop 2	ICMP	0.000	Successful
Laptop 2	Laptop 2	ICMP	0.000	Successful
PC 0	Laptop 1	ICMP	0.000	Successful

All three ICMP test returned a Successful status. Confirming that packets were correctly forwarded across multiple router hops. The routing time of 0.585 seconds for the PC0 to Laptop1 test reflects the multi-hop path traversed via the serial WAN links between Router0, Router2 and the 13.0.0.0 LAN segment.

❏ Conclusion: The Static routing LAB experiments was completed successfully. A multi-router topology comprising three Cisco routers interconnected via serial WAN links was designed and implemented. Interface IP addresses were assigned across eight network segments and static routes were ~~also~~ manually configured on each router - to provide full inter-network connectivity.